

Inside the network:

Hubs, modules, & motifs

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Recall: Hubs & bottlenecks

- **Hub:** high-Degree node; many direct interaction partners in the network.
- **Bottleneck:** high-Betweenness-Centrality node; sits on many shortest paths, regardless of its Degree.
- A protein can be a bottleneck without being a hub, and vice versa.

	High degree	Low degree
High betweenness	Hub-bottleneck	Bottleneck
Low betweenness	Hub	Regular node

Modules

- **Module (community/cluster):** a group of nodes more densely connected to each other than to the rest of the network
- Biologically, modules often correspond to protein complexes, functional pathways, or co-regulated groups
- A network's global topology tells you it's scale-free; module detection tells you which specific proteins belong together functionally

Importance of modules

- **Disease modules:** disease-associated genes tend to cluster together in the interactome rather than scatter randomly
- A module gives you a focused, biologically coherent shortlist instead of hundreds of loosely related hits
- Sub-modules within a larger module can point to distinct sub-complexes or regulatory sub-units
- Modules are the natural unit for downstream pathway enrichment analysis (coming in a later session)

MCODE: Molecular Complex Detection

- A Cytoscape plug-in that automatically detects densely-connected regions (complexes/modules) in a network.
- Originally designed to predict protein complexes from PPI data.
- Works in three stages: vertex weighting → complex prediction (seeding + expansion) → optional post-processing (fluff/hairecut)
- Outputs one or more sub-networks, each with an MCODE score reflecting how densely connected it is

MCODE algorithm: Vertex weighting

- Every node is scored using its local neighbourhood density, via the k-core of its immediate neighbourhood
- **k-core:** the largest sub-graph in which every node has at least degree k; a measure of how tightly a local region is interconnected
- **A node's weight = (highest k-core of its neighbourhood) × (density of that k-core)**
- **Result:** nodes embedded in tight local clusters score higher than nodes with many loosely-connected neighbours

MCODE algorithm: Seeding & expansion

- The highest-weighted node becomes a “seed” for a candidate complex
- The algorithm moves outward from the seed, recursively including neighbouring nodes whose weight is above a threshold (a fraction of the seed's own weight)
- Expansion stops once no more qualifying neighbours are found; this defines the boundary of the module
- The process repeats with the next unassigned highest-weighted node, producing multiple ranked modules from one network

MCODE parameters

- **Degree cutoff:** minimum degree a node must have to even be considered (default 2); filters out peripheral nodes early.
- **Node score cutoff:** minimum vertex-weight percentage (relative to the seed) required to join a growing cluster (default 0.2).
- **K-core:** minimum k-core value required within the final complex (default 2) — enforces internal density on the output

MCODE parameters

- **Max depth:** how many steps outward from the seed the expansion is allowed to search (default 100)
- **Haircut:** prunes singly-connected nodes (degree-1 members) from the final complex
- **Fluff:** optionally adds one extra layer of loosely-connected neighbours around the core, controlled by a fluff density threshold

Interpreting MCODE output

- Each detected cluster gets an MCODE score = (k-core level) × (number of nodes) — higher score → denser, larger module
- Clusters are ranked; Cluster 1 is the densest/most significant complex found.
- Each cluster can be extracted as its own sub-network for separate analysis (File → New Network → From Selected Nodes).
- A high-scoring module with no known shared function is itself a hypothesis: these proteins likely work together.

Session task 1

1. Select STRING disease query from network search bar
2. Search for Pancreatic cancer
3. Run network analyzer as taught in the previous session and identify the hub protein
4. Select MCODE from apps
5. Run the analysis with default parameters
6. Check the proteins by hub degree, is your hub still present?

CytoHubba

- cytoHubba is a Cytoscape plug-in that computes 11 different ranking methods and lets you cross-validate which nodes are consistently important across several of them
- Different centrality metrics can disagree: a node ranked #1 by Degree may rank poorly by Betweenness, or vice versa
- A gene ranked highly across multiple independent methods is a much stronger hub/target candidate than one ranked highly by only one

CytoHubba | Local methods

- **Maximum Neighborhood Component (MNC):** size of the largest connected component within a node's immediate neighbourhood. It filters out neighbours that don't actually connect to each other.
- **Density of MNC (DMNC):** MNC size normalised by neighbourhood size, favouring nodes whose neighbourhood is small but tightly interconnected.
- These look only at each node's immediate surroundings. Fast to compute, but blind to a node's role in the network as a whole.

CytoHubba | Global methods

- **Radiality:** how central a node is relative to the network's diameter.
High radiality means short paths to everywhere, scaled by overall network size.
- **Stress:** Counts how many shortest paths pass through a node, without normalising by total path count (an un-normalised cousin of Betweenness). Nodes with high stress are heavily “trafficked” chokepoints.
- **Eccentricity:** same concept as in Network Analyzer: worst-case distance to any other node.

CytoHubba | Hybrid methods

- **Maximal Clique Centrality (MCC):** sums the geometric-mean scores of all maximal cliques containing a node. Rewards nodes embedded in tight, fully-connected sub-groups. Most important metric
- **Edge Percolated Component (EPC):** estimates how well a node maintains network connectivity under simulated random edge removal (percolation); a robustness-based importance score.

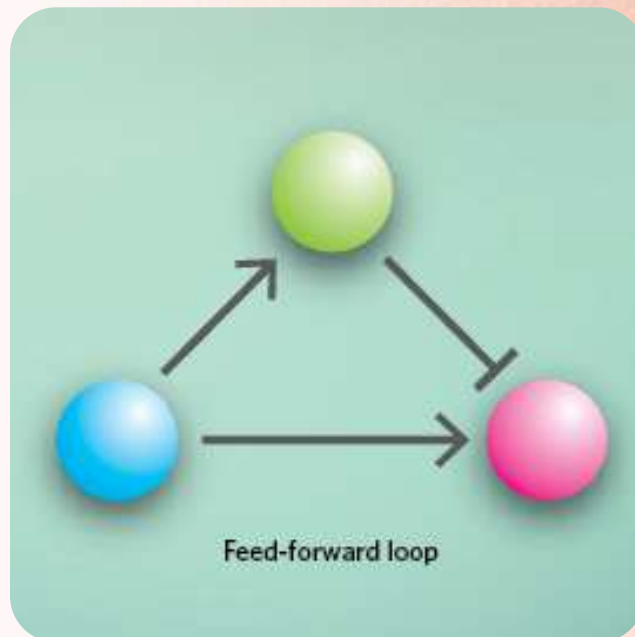
Session task 2

1. On the same pancreatic cancer network, select CytoHubba from the side panel
2. Run the analysis for all 11 methods
3. Select top 20 genes for each of: degree, MCC, and bottleneck
4. Lay the three top 20 lists side by side, and mark all genes present in the three lists.

Network motifs

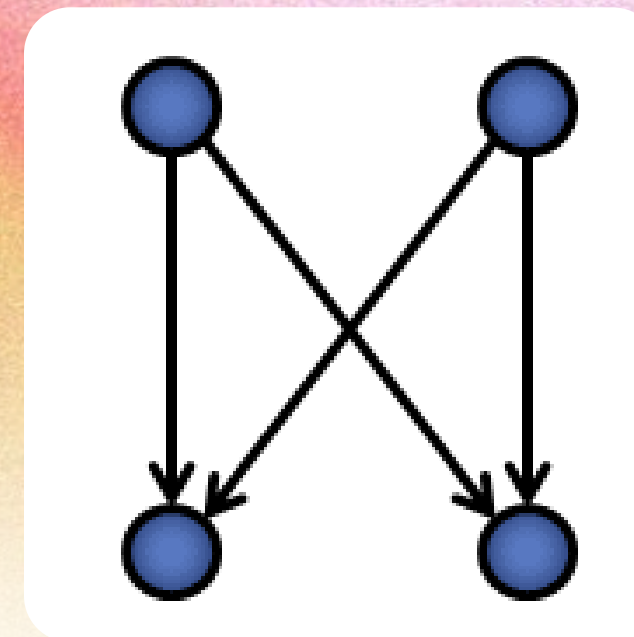
- **Motif:** a small, recurring pattern of connections (typically 3–4 nodes) that appears far more often in the real network than in a randomised network with the same degree distribution.
- Motifs are the network equivalent of a recurring circuit design: the same small wiring pattern re-used wherever a particular regulatory logic is needed
- Statistical significance matters: a pattern only counts as a motif if it's over-represented relative to comparable random networks, not just present
- Motif finding requires directed edges, which is not the case for many PPIs

Common motif types



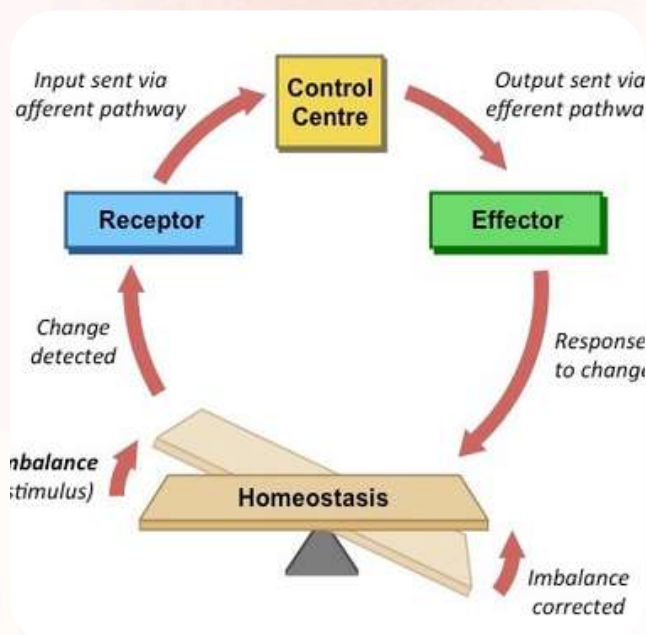
Feed-forward loop (FFL)

- $X \rightarrow Y \rightarrow Z$ and $X \rightarrow Z$ (two parallel paths to Z)
- Signal processing unit



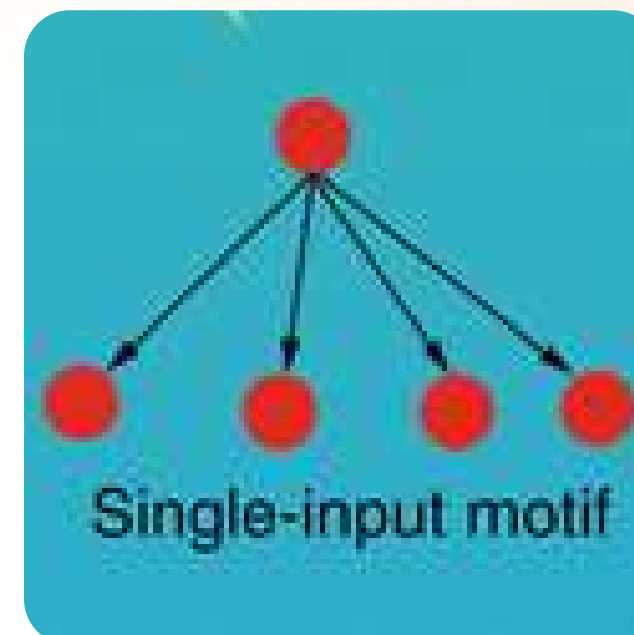
Bi-fan

- Two inputs $\rightarrow$ two outputs ($X1, X2 \rightarrow Z1, Z2$)
- Coordinated regulation



Feedback loop

- Closed cycle ($X \rightarrow Y \rightarrow X$ or longer)
- Negative: stability/oscillations; Positive: bistability/switches



Single-input motif

- One master regulator $\rightarrow$ multiple targets ($X \rightarrow Z1, Z2, \dots Zn$)
- Pathway coordination

Project task 2

1. Run CytoHubba analysis on your disease's network as we did in this session (MCC, bottleneck, hub)
2. Identify Hub-bottleneck proteins
3. Discuss the proteins you identified and their significance in your disease using available literature

Deliverables:

1. The three top 20 lists
2. A list of common proteins
3. A document discussing the proteins you found and their significance in your disease

THANK YOU!

Any questions?